Chapter 5:
Limits and Continuity

5.1 Limits

We now want to define the limit of a function. In Calculus you might have learned that

$$\lim_{x \to 3} 2x = 6$$

since

$\boldsymbol{x}$	$\boldsymbol{f(x) = 2x}$
4	8
3.5	7
3.1	6.2
3.01	6.02
3.001	6.002

Example. Define $f: \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

What is

$$\lim_{x \to \sqrt{2}} f(x) = ?$$

$\boldsymbol{x}$	$\boldsymbol{f(x)}$
1	1
1.4	1
1.41	1
1.414	1
1.4142	1

Hence we might suspect

$$\lim_{x \to \sqrt{2}} f(x) = 1$$

Example. Define $f: \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

What is

$$\lim_{x \to \sqrt{2}} f(x) = ?$$

$\boldsymbol{x}$	$\boldsymbol{f(x)}$
$\sqrt{2} - 0.1$	0
$\sqrt{2} - 0.01$	0
$\sqrt{2} - 0.001$	0
$\sqrt{2} - 0.0001$	0
$\sqrt{2} - 0.00001$	0

Depending on which sequence of values we choose for x, we seem to compute that the limit is 0 or 1.

Definition 4.7. *Let (f_n) be a sequence and $L \in \mathbb{R}$.*

We say that (f_n) converges to L when:

For every $\epsilon > 0$ there exists $M \in \mathbb{N}^+$ such that if $n > M$ then $|f_n - L| < \epsilon$.

Conceptually

For every $\epsilon > 0$ if n is close enough to ∞, i.e. n is large, then $|f_n - L| < \epsilon$.

We can similarly describe the limit of a function $f(x)$ as $x \to a$ by

For every $\epsilon > 0$ if x is close enough to a, then $|f(x) - L| < \epsilon$.

By "x is close to a" we mean that for some $\delta > 0$ we have the distance of x to a is $< \delta$, i.e. $|x - a| < \delta$.

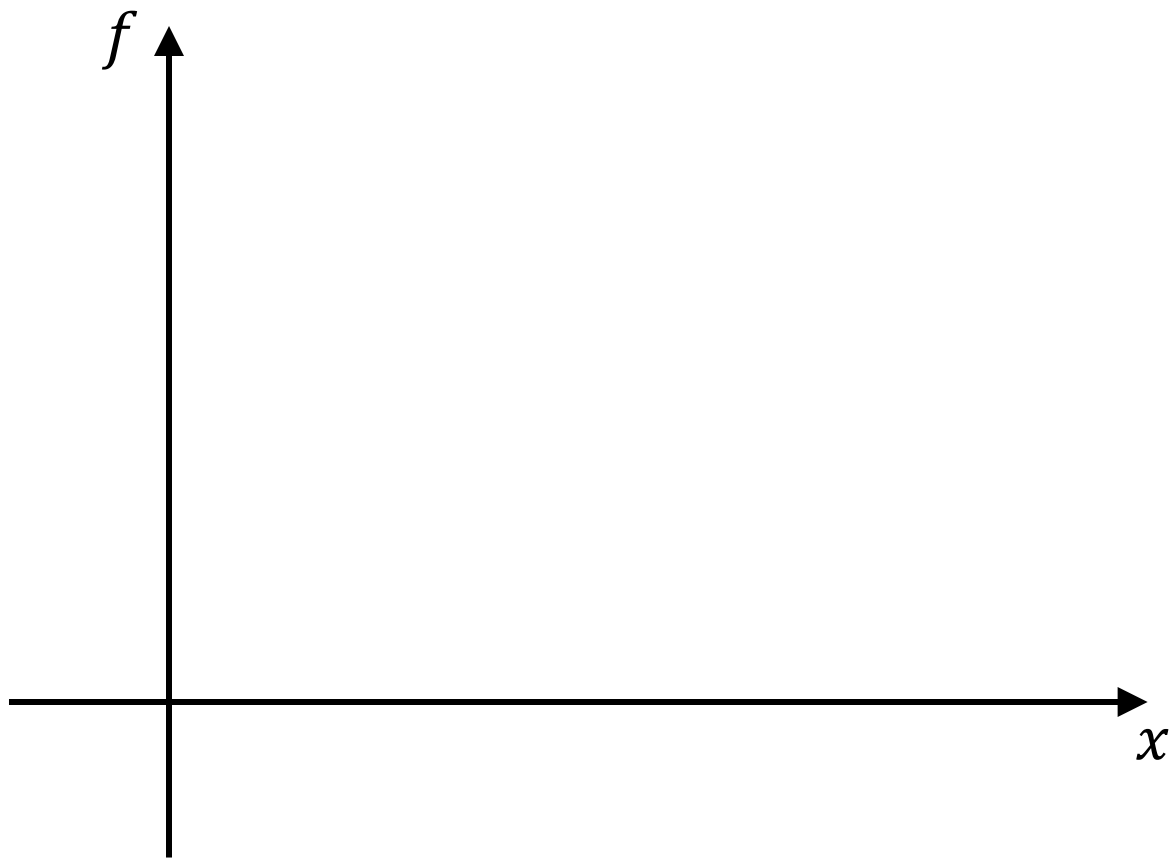
$\longrightarrow$

Proposed Definition. *Let* $f: \mathbb{R} \to \mathbb{R}$ *be function,* $a \in \mathbb{R}$ *and* $L \in \mathbb{R}$. *We write*

$$\lim_{x \to a} f(x) = L$$

when

for all $\epsilon > 0$ *there exists* $\delta > 0$ *such that if* $0 < |x - a| < \delta$ *then* $|f(x) - L| < \epsilon$.



Example. Prove

$$\lim_{x \to 3} 2x = 6$$

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Scrap work:

Proof:

Example. Consider

$$\lim_{x \to 9} \sqrt{x} = 3$$

The domain of $\sqrt{x}$ is $\{ x \in \mathbb{R} \mid x \geq 0 \}$, but we have only defined limits a functions whose domain is $\mathbb{R}$.

Proposed Definition 2. *Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ be function, $a \in \mathbb{R}$ and $L \in \mathbb{R}$. We write*

$$\lim_{x \to a} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in E$ and $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$.

For which values of a should we consider limits as $x \to a$?

Example. Consider

$$\lim_{x \to -5} \sqrt{x}.$$

This is nonsense since $\sqrt{x}$ is not defined for any x near -5.

Maybe we should require $a \in E$, i.e. a is in the domain of f…

Example. Consider

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2}.$$

This limit should make sense but $x = 2$ is not in the domain of $\frac{x^2-4}{x-2}$.

Definition 5.1. *Let $E \subseteq \mathbb{R}$ and $a \in \mathbb{R}$. We say that a is a* <u>*limit point*</u> *of E if for every $\delta > 0$ there exists a point $x \in E$ such that $0 < |x - a| < \delta$.*

Theorem. *Let $E \subseteq \mathbb{R}$ and $a \in \mathbb{R}$. a is a limit point of E if and only if there exists a sequence* (x_n) of points such that $x_n \in E$, $x_n \neq a$, and $x_n \to a$.

Example. Consider the function $f: (0, \infty) \to \mathbb{R}$ defined by $f(x) = \log x$.

The domain of f is $E = (0, \infty)$.

The set of limit points of $E = (0, \infty)$ is

Example. Consider the set $E = \mathbb{N}$ of all natural numbers.

The set of limit points of $E = \mathbb{N}$ is

Official Definition 5.2. *Let $E \subseteq \mathbb{R}$, a be a limit point of E, $f: E \to \mathbb{R}$ be function, and $L \in \mathbb{R}$. We write*

$$\lim_{x \to a} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in E$ and $0 < |x - a| < \delta$ then $|f(x) - L| < \epsilon$.

Example. Show that

$$\lim_{x \to 2} \frac{x^2 - 4}{x - 2} = 4.$$

Scrap work:

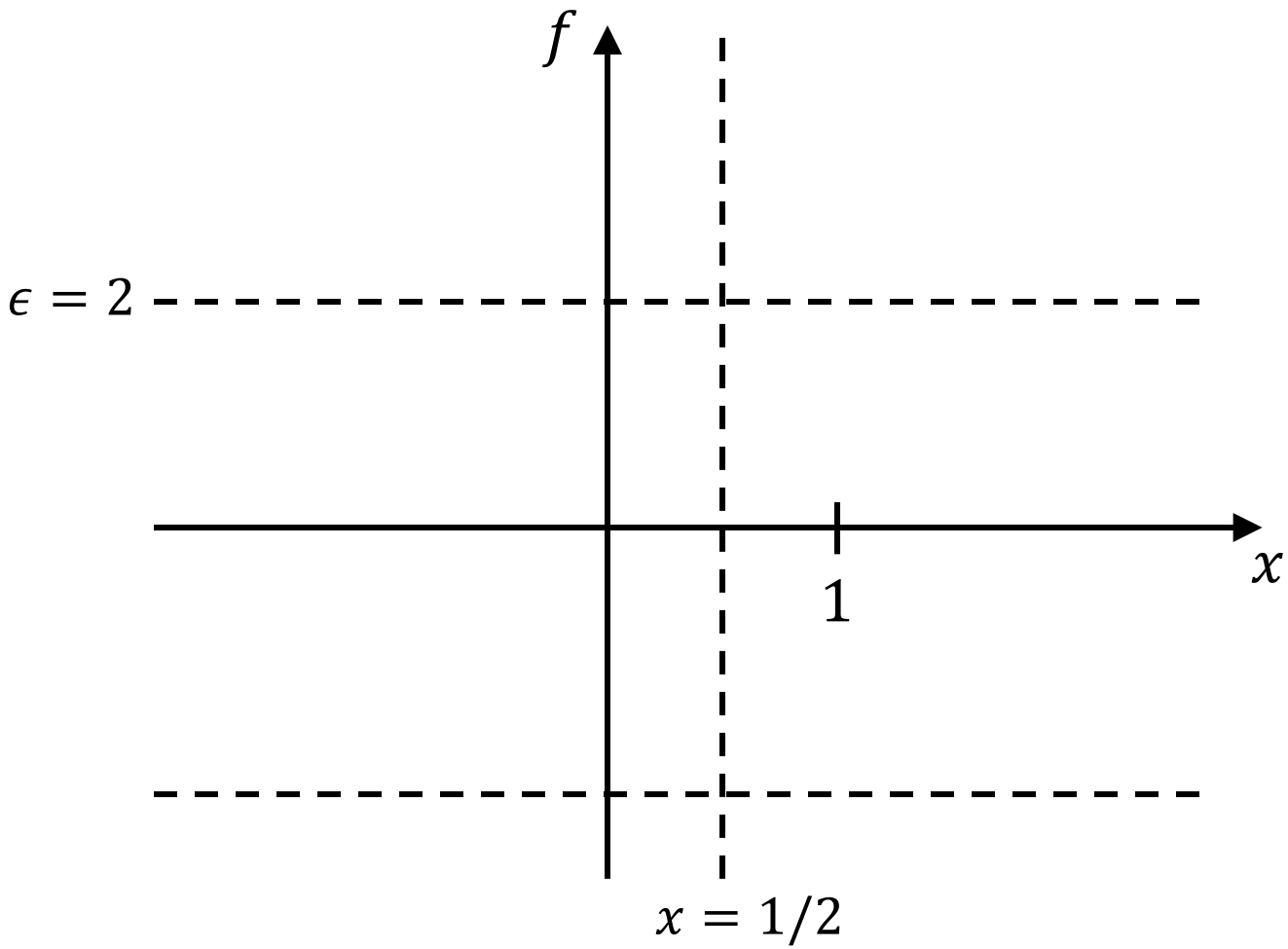
Proof:

Example. Show that

$$\lim_{x \to 1} \frac{x^2 - 1}{2x - 1} = 0.$$

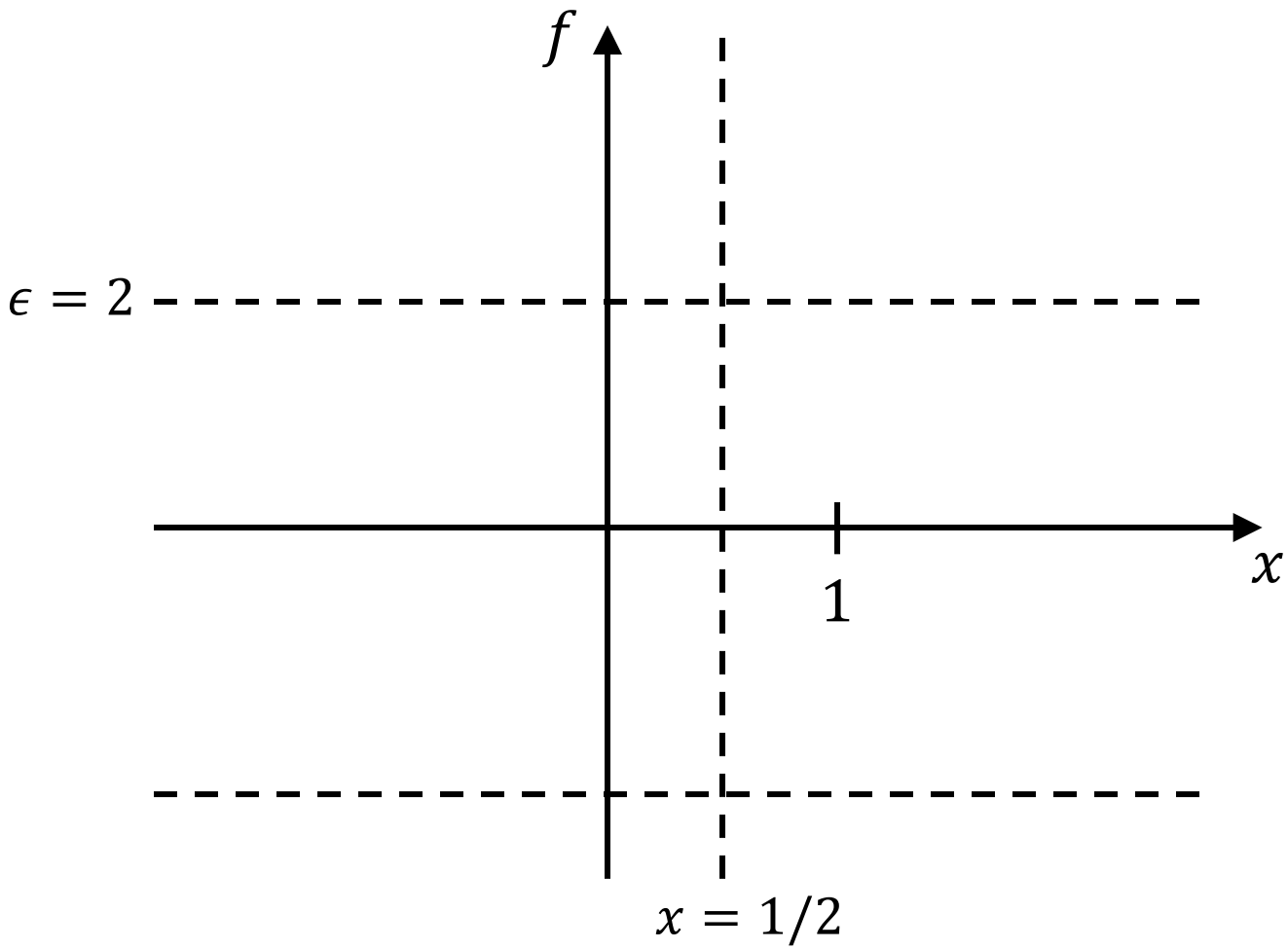
Example. Show that

$$\lim_{x \to 1} \frac{x^2 - 1}{2x - 1} = 0.$$



Example. Show that

$$\lim_{x \to 1} \frac{x^2 - 1}{2x - 1} = 0.$$



Example. Show that

$$\lim_{x \to 1} \frac{x^2 - 1}{2x - 1} = 0.$$

Theorem 5.3. (Algebra of Limits Theorem)

Let $E \subseteq \mathbb{R}$, *let* a *be a limit point of* E, *and* $f, g: E \to \mathbb{R}$. *If*

$$\lim_{x \to a} f(x) = \alpha \qquad \text{and} \qquad \lim_{x \to a} g(x) = \beta$$

then

i. $\lim_{x \to a} (f(x) + g(x)) = \alpha + \beta$;

ii. $\lim_{x \to a} f(x) \cdot g(x) = \alpha \cdot \beta$;

iii. $\lim_{x \to a} f(x)/g(x) = \alpha/\beta$ (*provided* $\beta \neq 0$)

Theorem. *Let $E \subseteq \mathbb{R}$, let a be a limit point of E, and $f, g: E \to \mathbb{R}$. Assume that $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ both exist.*

- *If $f(x) \leq g(x)$ for all $x \in E$, then*

$$\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x).$$

- *If $f(x) \leq c$ for all $x \in E$, then*

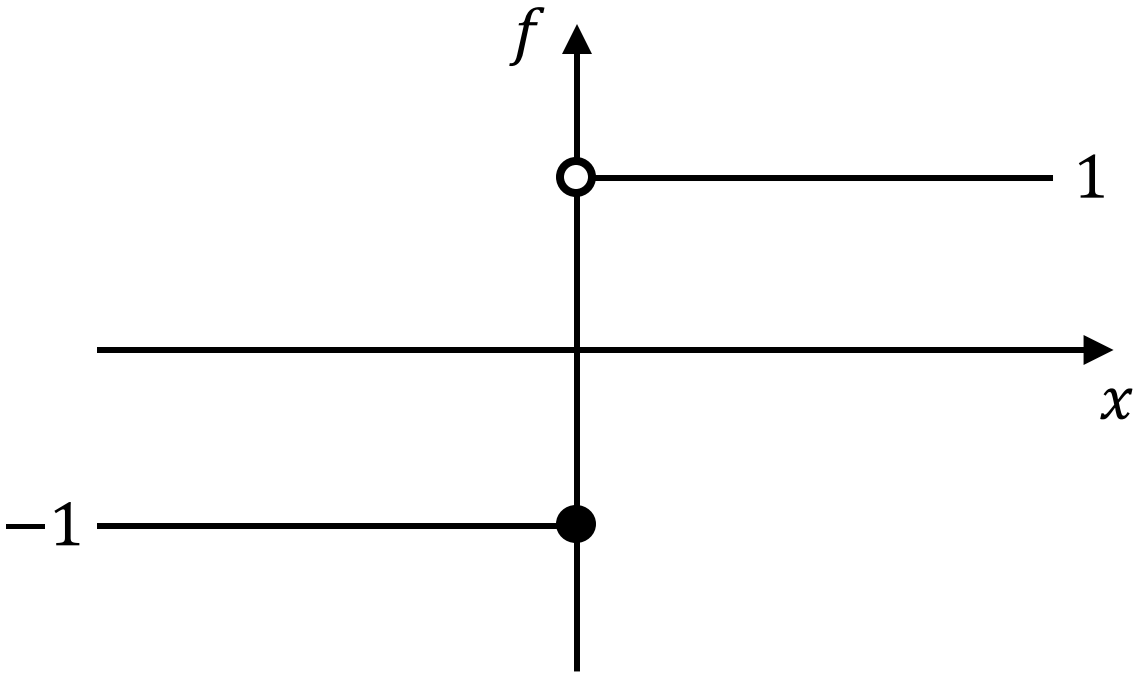
$$\lim_{x \to a} f(x) \leq c.$$

- *If $b \leq f(x)$ for all $x \in E$, then*

$$b \leq \lim_{x \to a} f(x).$$

Example. Define $f: \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x \leq 0 \end{cases}$$



Definition 5.4. *Let $E \subseteq \mathbb{R}$, a be a limit point of E and $f: E \to \mathbb{R}$ be function. We say $f(x)$ <u>diverges</u> as x approaches a if*

$$\lim_{x \to a} f(x) = L$$

is false for all $L \in \mathbb{R}$.

Looking back at our definition of limit, let us focus on:

$$0 < |x - a| < \delta$$

The left-side gives us:

The right-side gives us:

Definition 5.5. *Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ be function and $L \in \mathbb{R}$. Suppose that a is a limit point of $E \cap (a, \infty)$. We write*

$$\lim_{x \to a^+} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in E$ and $a < x < a + \delta$ then $|f(x) - L| < \epsilon$.

Definition 5.6. *Let $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ be function and $L \in \mathbb{R}$. Suppose that a is a limit point of $E \cap (-\infty, a)$. We write*

$$\lim_{x \to a^-} f(x) = L$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that if $x \in E$ and $a - \delta < x < a$ then $|f(x) - L| < \epsilon$.

Theorem 5.7. *Let* $E \subseteq \mathbb{R}$, $f: E \to \mathbb{R}$ *be function and* $L \in \mathbb{R}$. *Suppose that* a *is a limit point of both* $E \cap (-\infty, a)$ *and* $E \cap (a, \infty)$. *Then*

$$\lim_{x \to a} f(x) = L$$

if and only if

$$\lim_{x \to a^-} f(x) = L \qquad \text{and} \qquad \lim_{x \to a^+} f(x) = L.$$

Definition 5.9. *Let $E \subseteq \mathbb{R}$, a be a limit point of E, $f: E \to \mathbb{R}$ a function. We write*

$$\lim_{x \to a} f(x) = \infty$$

if for every $r \in \mathbb{R}$ there exists $\delta > 0$ such that if $x \in E$ and $0 < |x - a| < \delta$ then $f(x) > r$.

Example. Show that

$$\lim_{x \to 0} \frac{1}{x^2} = \infty$$

Scrap work:

Proof:

Definition 5.10. *Let $E \subseteq \mathbb{R}$ be a set which is not bounded above. Let $f: E \to \mathbb{R}$ be function and $L \in \mathbb{R}$. We write*

$$\lim_{x \to \infty} f(x) = L$$

if for every $\epsilon > 0$ there exists $M \in \mathbb{R}$ s.t. if $x \in E$ and $x > M$ then $|f(x) - L| < \epsilon$.

Definition 5.11. *Let $E \subseteq \mathbb{R}$ be a set which is not bounded above. Let $f: E \to \mathbb{R}$ be function. We say $f(x)$ <u>diverges</u> as x approaches infinity if*

$$\lim_{x \to \infty} f(x) = L$$

is false for all $L \in \mathbb{R}$.

Definition 5.12. *Let $E \subseteq \mathbb{R}$ be a set which is not bounded above. Let $f: E \to \mathbb{R}$ be a function. We write*

$$\lim_{x \to \infty} f(x) = \infty$$

if for every $r \in \mathbb{R}$ there exists $M \in \mathbb{R}$ s.t. if $x \in E$ and $x > M$ then $f(x) > r$.

Example. Show that $\sin x$ diverges as x approaches infinity.

Example. Show that

$$\lim_{x \to \infty} (2x - 1) = \infty$$